Step 2: Divide the mass of each element by its atomic mass. This gives the relative number of moles of various elements in the compound.

Step 3: Divide the value of relative number of moles obtained in the step 2 by the smallest number of them to get the simplest ratio.

Step 4: (only if necessary) in case the simplest ratios obtained in the step 3 are not whole numbers then they may be converted into whole number by multiplying by a suitable smallest number.

Example:

1. An acid found in Tamarind on analysis shows the following percentage composition: 32 % Carbon; 4 % Hydrogen; 64 % Oxygen. Find the empirical formula of the compound.

Element	Percentage	molar mass	Relative no. of moles	Simplest ratio	Simplest ratio (in whole nos)
C	32	12	$\frac{32}{12} = 2.66$	$\frac{2.66}{2.66} = 1$	2
H	4	1	$\frac{4}{1} = 4$	$\frac{4}{2.66} = 1.5$	3
O	64	16	$\frac{64}{16} = 4$	$\frac{4}{2.66} = 1.5$	3

The empirical formula is $C_2H_3O_3$

2. An organic compound present in vinegar has 40 % carbon, 6.6 % hydrogen and 53.4 % oxygen. Find the empirical formula of the compound.

Element	Percentage	Atomic Mass	Relative no. of moles	Simplest ratio	Simplest ratio (in whole no)
C	40	12	$\frac{40}{12} = 3.3$	$\frac{3.3}{3.3} = 1$	1
H	6.6	1	$\frac{6.6}{1} = 6.6$	$\frac{6.6}{3.3} = 2$	2
O	53.4	16	$\frac{53.4}{16} = 3.3$	$\frac{3.3}{3.3} = 1$	1

The empirical formula is CH_2O

Evaluate Yourself

5) A Compound on analysis gave the following percentage composition C=54.55%, H=9.09%, O=36.36%. Determine the empirical formula of the compound.

Molecular formula of a compound is a whole number multiple of the empirical formula. The whole number can be calculated from the molar mass of the compound using the following expression

$$\text{Whole number (n)} = \frac{\text{Molar mass of the compound}}{\text{Calculated empirical formula mass}}$$

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